



Virtual Internship

Normalizing Denormalized Order Data

Task 2

01

Understand the concept

Before transforming. Here is what denormalized and normalized actually mean and why the difference matters

fact_orders_line.csv | MS Excel | Power Query |

What is Denormalized Data ?

Everything is stored flat, in a single table, Customer details, Product details , Order details and all repeat on every single row

Denormalized Table

	A	B	C	D	E	F	G	H	I	J	K	L
1	order_id	order_placement_date	mmm_yy	week_no	customer_id	customer_name	city	product_id	product_name	category	order_qty	delivery_qty
2	FMR34203601	01-03-2022 Mar_22	W 10		789203	Rel Fresh	Vadodara	25891601	AM Tea 500	beverages	110	110
3	FMR34220601	01-03-2022 Mar_22	W 10		789220	Atlas Stores	Surat	25891601	AM Tea 500	beverages	78	78
4	FMR32501601	01-03-2022 Mar_22	W 10		789501	Viveks Stores	Surat	25891601	AM Tea 500	beverages	139	139
5	FMR33102601	01-03-2022 Mar_22	W 10		789102	Vijay Stores	Ahmedabad	25891601	AM Tea 500	beverages	180	180
6	FMR32902601	01-03-2022 Mar_22	W 10		789902	Elite Mart	Ahmedabad	25891601	AM Tea 500	beverages	93	88
7	FMR33903601	01-03-2022 Mar_22	W 10		789903	Elite Mart	Vadodara	25891601	AM Tea 500	beverages	93	84
8	FMR32421601	01-03-2022 Mar_22	W 10		789421	Lotus Mart	Ahmedabad	25891601	AM Tea 500	beverages	108	103
9	FMR34421601	01-03-2022 Mar_22	W 10		789421	Lotus Mart	Ahmedabad	25891601	AM Tea 500	beverages	75	75
10	FMR34621601	01-03-2022 Mar_22	W 10		789621	Expert Mart	Ahmedabad	25891601	AM Tea 500	beverages	170	170
11	FMR33520601	01-03-2022 Mar_22	W 10		789520	Acclaimed Stores	Surat	25891601	AM Tea 500	beverages	182	146
12	FMR32702601	01-03-2022 Mar_22	W 10		789702	Sorefoz Mart	Ahmedabad	25891601	AM Tea 500	beverages	93	74
13	FMR34122601	01-03-2022 Mar_22	W 10		789122	Coolblue	Vadodara	25891601	AM Tea 500	beverages	68	61
14	FMR32101601	01-03-2022 Mar_22	W 10		789101	Vijay Stores	Surat	25891601	AM Tea 500	beverages	124	124
15	FMR33103602	01-03-2022 Mar_22	W 10		789103	Vijay Stores	Vadodara	25891601	AM Tea 500	beverages	188	150
16	FMR32202603	01-03-2022 Mar_22	W 10		789202	Rel Fresh	Ahmedabad	25891601	AM Tea 500	beverages	125	119
17	FMR34522601	01-03-2022 Mar_22	W 10		789522	Acclaimed Stores	Vadodara	25891601	AM Tea 500	beverages	181	163
18	FMR33321601	01-03-2022 Mar_22	W 10		789321	Chiptec Stores	Ahmedabad	25891601	AM Tea 500	beverages	157	157
19	FMR33201602	01-03-2022 Mar_22	W 10		789201	Rel Fresh	Surat	25891601	AM Tea 500	beverages	180	171
20	FMR34622602	01-03-2022 Mar_22	W 10		789622	Expert Mart	Vadodara	25891601	AM Tea 500	beverages	96	91
21	FMR34203601	01-03-2022 Mar_22	W 10		789203	Rel Fresh	Vadodara	25891301	AM Ghee 250	Dairy	100	100
22	FMR33320501	01-03-2022 Mar_22	W 10		789320	Chiptec Stores	Surat	25891301	AM Ghee 250	Dairy	46	46
23	FMR33220603	01-03-2022 Mar_22	W 10		789220	Atlas Stores	Surat	25891301	AM Ghee 250	Dairy	75	75
24	FMR33721603	01-03-2022 Mar_22	W 10		789721	Logic Stores	Ahmedabad	25891301	AM Ghee 250	Dairy	59	59
25	FMR32420403	01-03-2022 Mar_22	W 10		789420	Lotus Mart	Surat	25891301	AM Ghee 250	Dairy	87	87
26	FMR33420301	01-03-2022 Mar_22	W 10		789420	Lotus Mart	Surat	25891301	AM Ghee 250	Dairy	23	18
27	FMR32121301	01-03-2022 Mar_22	W 10		789121	Coolblue	Ahmedabad	25891301	AM Ghee 250	Dairy	72	65
28	FMR33421301	01-03-2022 Mar_22	W 10		789421	Lotus Mart	Ahmedabad	25891301	AM Ghee 250	Dairy	75	67
29	FMR34421301	01-03-2022 Mar_22	W 10		789421	Lotus Mart	Ahmedabad	25891301	AM Ghee 250	Dairy	47	45
30	FMR33402303	01-03-2022 Mar_22	W 10		789402	Propel Mart	Ahmedabad	25891301	AM Ghee 250	Dairy	94	94
31	FMR34520301	01-03-2022 Mar_22	W 10		789520	Acclaimed Stores	Surat	25891301	AM Ghee 250	Dairy	38	34
32	FMR32422301	01-03-2022 Mar_22	W 10		789422	Lotus Mart	Vadodara	25891301	AM Ghee 250	Dairy	63	63
33	FMR33422301	01-03-2022 Mar_22	W 10		789422	Lotus Mart	Vadodara	25891301	AM Ghee 250	Dairy	63	60
34	FMR33301301	01-03-2022 Mar_22	W 10		789301	Expression Stores	Surat	25891301	AM Ghee 250	Dairy	92	83

What is Normalized Data ?

Same table is splitted into fact tables (The transaction) and dimension tables (the entities) connected by keys

- 1 No Redundancy
- 2 Single Source of Truth
- 3 Cleaner Relationship

Normalized Table

Fact_orders Table

1	order_id	order_placement_date	customer_id	product_id	order_qty	delivery_qty
2	FAP411221601	10-04-2022	789221	25891601	188	188
3	FAP412221601	31-03-2022	789221	25891601	200	200
4	FAP412221601	09-04-2022	789221	25891601	183	183
5	FAP416221601	15-04-2022	789221	25891601	78	74
6	FAP417221601	16-04-2022	789221	25891601	57	57
7	FAP418221601	17-04-2022	789221	25891601	180	180
8	FAP419221601	16-04-2022	789221	25891601	59	59
9	FAP422221601	20-04-2022	789221	25891601	101	101
10	FAP424221601	23-04-2022	789221	25891601	117	105
11	FAP425221601	23-04-2022	789221	25891601	178	178
12	FAP428221602	26-04-2022	789221	25891601	175	175
13	FAP429221601	28-04-2022	789221	25891601	177	177
14	FAP43221601	02-04-2022	789221	25891601	149	134
15	FAP44221601	01-04-2022	789221	25891601	185	148

Dim_date Table

1	order_placement_date	mmm_yy	week_no
2	01-03-2022	Mar_22	W 10
3	02-03-2022	Mar_22	W 10
4	03-03-2022	Mar_22	W 10
5	04-03-2022	Mar_22	W 10
6	05-03-2022	Mar_22	W 10
7	06-03-2022	Mar_22	W 11
8	07-03-2022	Mar_22	W 11
9	08-03-2022	Mar_22	W 11
10	09-03-2022	Mar_22	W 11
11	10-03-2022	Mar_22	W 11
12	11-03-2022	Mar_22	W 11
13	12-03-2022	Mar_22	W 11
14	13-03-2022	Mar_22	W 12
15	14-03-2022	Mar_22	W 12

Dim_product Table

1	product_id	product_name	category
2	25891101	AM Milk 500	Dairy
3	25891102	AM Milk 250	Dairy
4	25891103	AM Milk 100	Dairy
5	25891201	AM Butter 100	Dairy
6	25891202	AM Butter 250	Dairy
7	25891203	AM Butter 500	Dairy
8	25891301	AM Ghee 250	Dairy
9	25891302	AM Ghee 150	Dairy
10	25891303	AM Ghee 100	Dairy
11	25891401	AM Curd 250	Dairy
12	25891402	AM Curd 100	Dairy
13	25891403	AM Curd 50	Dairy
14	25891501	AM Biscuits 750	Food
15	25891502	AM Biscuits 500	Food

Dim_customer Table

1	customer_id	customer_name	city
2	789101	Vijay Stores	Surat
3	789102	Vijay Stores	Ahmedabad
4	789103	Vijay Stores	Vadodara
5	789121	Coolblue	Ahmedabad
6	789122	Coolblue	Vadodara
7	789201	Rel Fresh	Surat
8	789202	Rel Fresh	Ahmedabad
9	789203	Rel Fresh	Vadodara
10	789220	Atlas Stores	Surat
11	789221	Atlas Stores	Ahmedabad
12	789301	Expression Stores	Surat
13	789303	Expression Stores	Vadodara
14	789320	Chiptec Stores	Surat
15	789321	Chiptec Stores	Ahmedabad

02

Normalization Process

Built using Reference option not duplicate

| MS Excel | Power Query | Reference | Star Schema

1.Set up fact_order_lines as staging query and branch off with references

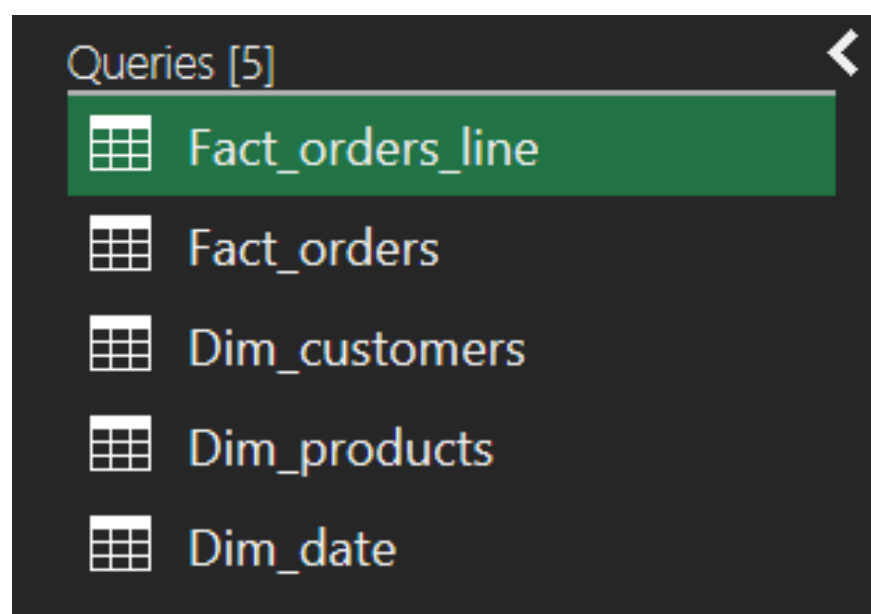
Problem

The fact_order_lines table holds customer, product and date fields all together one row per order line - denormalized

Approach

- Created 4 Reference > Fact_orders_line and renamed into
 - 1.Fact_orders
 - 2.Dim_customers
 - 3.Dim_products
 - 4.Dim_date

Result



2.Sort and remove duplicates by each table’s primary key

Fact_orders Table

APPLIED STEPS	
Source	
Removed Other Columns	⚙
Sorted Rows by order_id	
✕ Removed Duplicates by order_id	

Dim_customers Table

APPLIED STEPS	
Source	
Removed Other Columns	⚙
Sorted Rows by customer_id	
✕ Removed Duplicates by customer_id	

Dim_products Table

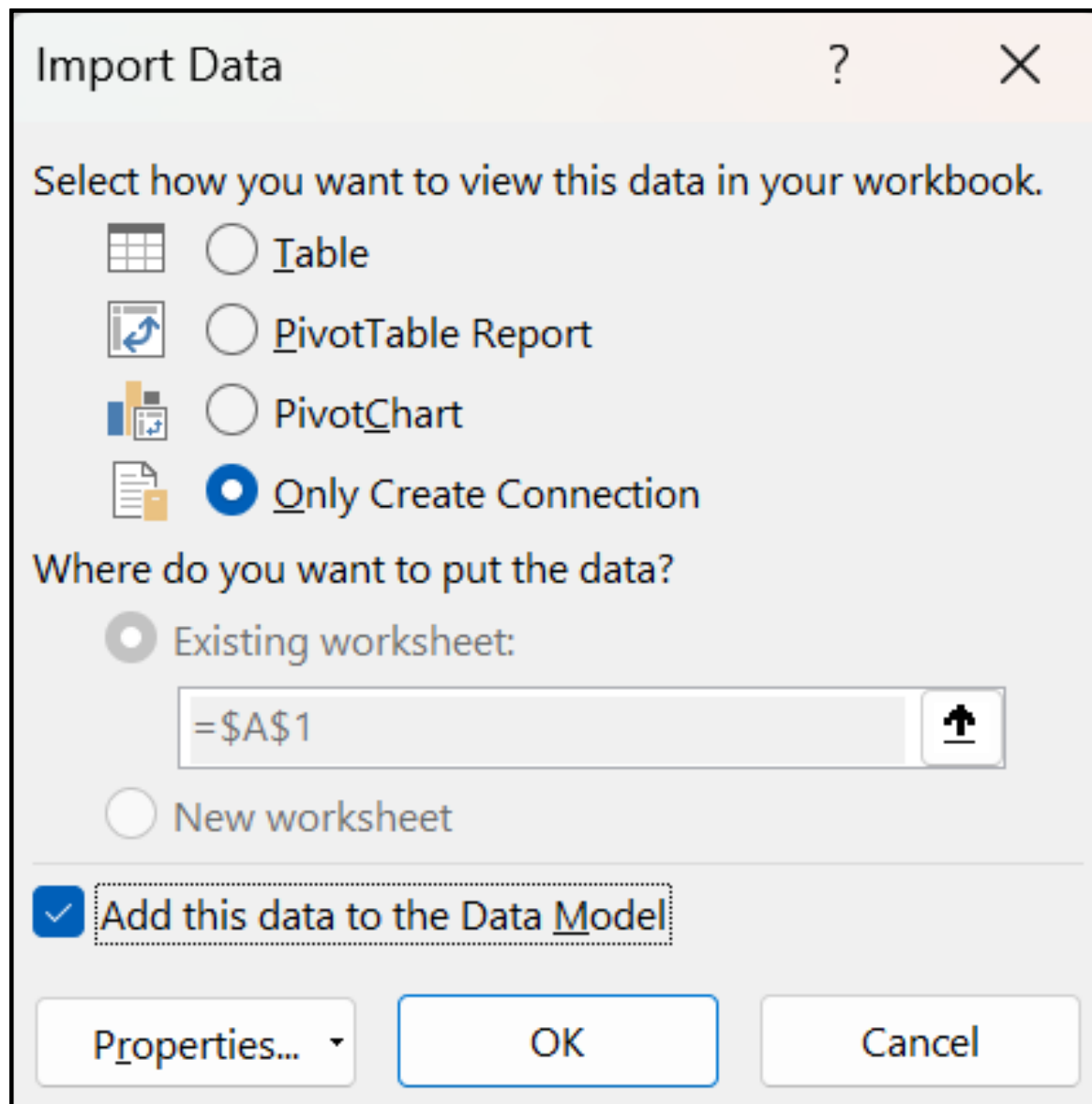
APPLIED STEPS	
Source	
Removed Other Columns	⚙
Sorted Rows by product_id	
✕ Removed Duplicates by product_id	

Dim_date Table

APPLIED STEPS	
Source	
Removed Other Columns	⚙
Sorted Rows by order_placement_date	
✕ Removed Duplicates by order_placement_date	

3.Load each query into the data model

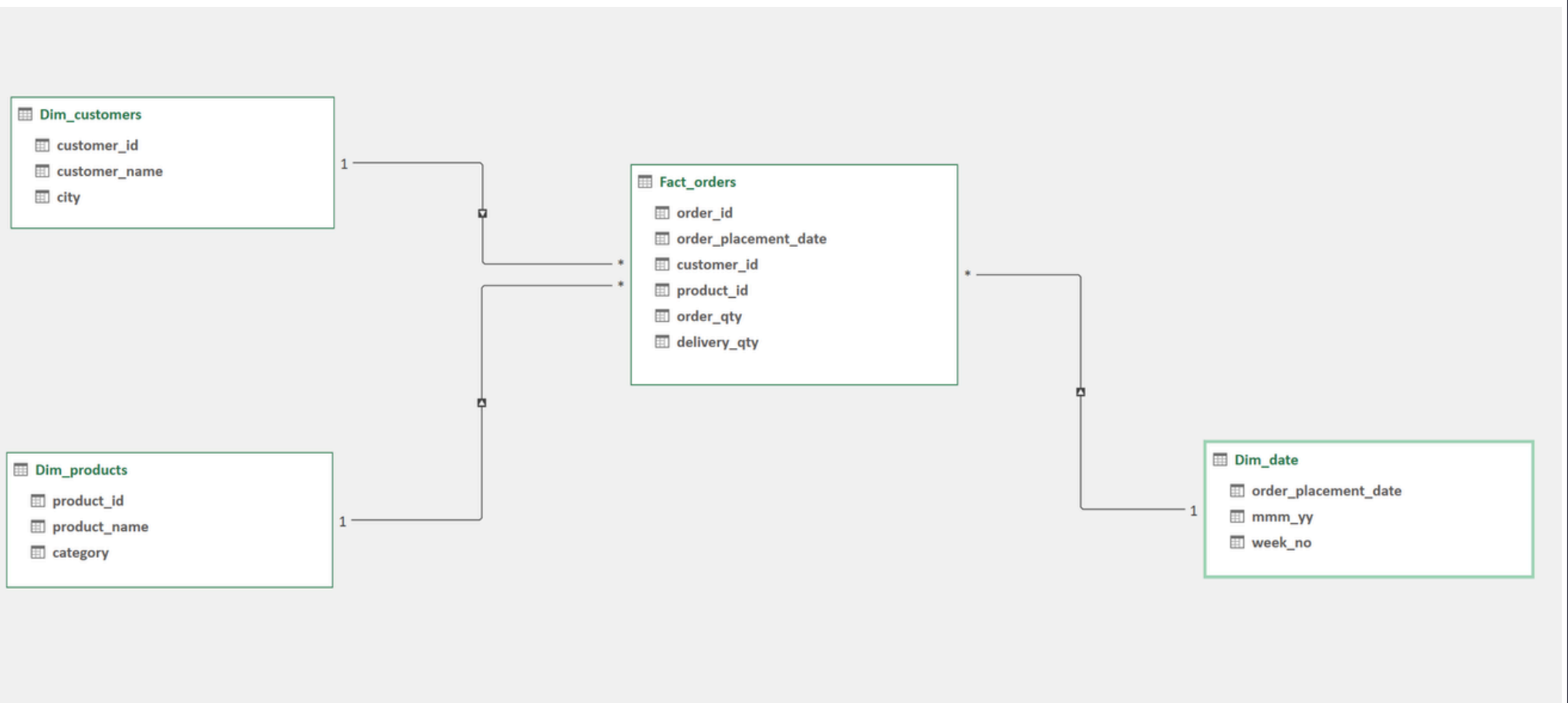
Approach



4.Build relationships in Diagram View

Approach

Power Pivot Ribbon > Manage > Diagram View > Connect each dimension to fact_orders through primary key



5.Choosen Reference over Duplicates

Reference

- No copy points to original
- Single Source of truth
- Fix once flows to all 4 automatically

Duplicate

- Full copy, frozen in time
- No link to original
- Fix a bug? Fix it 4 times

Row counts come out the same by using either approach

The task 2 didn't mention use duplicate or reference

This is my own call

Result

Row counts after normalization

 Fact_orders_line Connection only.
 Fact_orders 31,729 rows loaded. 
 Dim_customers 35 rows loaded.
 Dim_products 18 rows loaded.
 Dim_date 183 rows loaded.

236

Total rows across all three dimension tables combined
(183 + 35 + 18)

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Github Profile:

<https://github.com/Sachin-Analyst/AtliQ-Virtual-Internship-Week-1>